

BREAK INTO AEROSPACE

From Textile Engineering to Aerospace

Career Transition Guide

Targeted for the 2025–2026 Recruitment Cycle

What This Document Covers: Textile engineering is the discipline that most consistently surprises aerospace hiring managers — not because textile engineers lack relevant skills, but because the depth and specificity of those skills to the most technically demanding growth areas in aerospace are almost entirely invisible to both parties. The textile engineer applying to aerospace often does not know which of their specialisms are directly valuable. The aerospace recruiter who sees “textile engineering” on a CV does not immediately see composite preform design, fire-resistant cabin interior engineering, or technical textiles for protective systems. The result is a mutual failure of communication that leaves both sides underserved.

This guide is structured to make that communication possible — specifically to help the textile engineer identify which of their skills aerospace values, which gaps they need to close, and how to position their background in a language that aerospace interviewers recognise.

The honest starting point: textile engineering is a more niche transition than electrical, mechanical, computer science, or materials science engineering — the accessible role set is narrower and the direct technical overlap, while real and substantive, is concentrated in specific application domains rather than distributed across the sector. This does not make the transition less viable; it makes the targeting more important. A textile engineer who targets the right aerospace role — composite preform engineering, protective systems, cabin fire safety, spacecraft soft goods, or filtration systems — is not competing against aerospace engineers who understand the application better. They are competing as a domain specialist against engineers who may understand the aerospace context but who do not understand the textile science. That is a strong position. A textile engineer who applies diffusely to “aerospace structures” or “aerospace manufacturing” without the targeting is competing poorly against engineers whose foundational discipline is more directly aligned.

The aerospace roles genuinely accessible from a textile engineering background are: composite dry fibre preform engineering (the design and manufacture of woven, braided, and stitched reinforcement architectures for liquid composite moulding — the fastest-growing domain and the most direct application of textile expertise), cabin interior fire safety engineering (the flammability testing and certification of aircraft seat covers, carpets, curtains, and wall panels under FAR 25.853 and CS-25.853), emergency evacuation and protective systems engineering (inflatable evacuation slides, life rafts, restraint webbing, crew survival equipment — all large-scale technical textile products under strict airworthiness requirements), parachute and deceleration systems engineering (primarily defence and space, covering ejection seat parachutes, spacecraft recovery systems, and cargo delivery parachutes), spacecraft inflatable structure engineering (pressurised habitat modules, inflatable heat shields, airbag landing systems), space suit and pressure garment engineering (the most technically demanding soft goods application — multiple layers of technical textiles performing pressure containment, thermal regulation, and radiation protection simultaneously), filtration and fluid systems textiles (hydraulic and fuel system filters — porous textile structures performing separation under demanding pressure and chemical environments), and MLI and thermal blanket engineering (spacecraft multi-layer insulation using aluminised polymer films with textile scrim layers — a thermal management application where fabric architecture governs performance).

The employer landscape for these roles: composite preform manufacturers (Albany Composites, Safran Composites Preforms, Hexcel dry fabric division, Cytec/Solvay prepregs, A&P Technology braiding), OEM composite manufacturing groups (Airbus, Boeing, Embraer composite manufacturing — specifically their RTM and VARTM programmes), cabin interior system integrators (Safran Seats, Collins Aerospace Cabin Interiors, Recaro Aircraft Seating), protective systems manufacturers (Cobham Life Support, Survival Systems International, Eastern Aero Marine, EAM Worldwide), parachute and deceleration system specialists (Airborne Systems, Pioneer Aerospace, Irvin Aerospace), spacecraft soft goods specialists (ILC Dover — manufacturer of Apollo space suits and the BEAM habitat on the ISS, David Clark Company, Final Frontier Design), and spacecraft thermal hardware manufacturers (Sheldahl/Dunmore thermal films, Aerospace Fabrication and Materials).

PART I: WHAT AEROSPACE EMPLOYERS ACTUALLY SEE IN A TEXTILE ENGINEER

The Genuine Advantages — What Transfers Directly

The textile engineer's most directly transferable advantage is the understanding of fibre and fabric architecture as a structural design variable — an understanding that aerospace composite engineers frequently lack because they come from structural mechanics backgrounds where the reinforcement architecture is treated as a given rather than as a design space. The composite structural engineer knows classical laminate theory for UD tape laminates. The textile engineer knows how the crimp geometry of a woven fabric affects the in-plane stiffness relative to UD tape (the undulation of fibres through the weave thickness introduces a bending stress and reduces the effective fibre modulus), why a satin weave has better drapability than a plain weave (fewer crimp points per unit area, allowing the fabric to conform to curved tooling), why a braided preform provides better damage tolerance than a woven fabric (the interlacing architecture distributes impact energy through the thickness more effectively), and why a 3D woven preform eliminates the interlaminar weakness of laminated 2D composites (the z-fibres prevent delamination under out-of-plane loading). This architectural design knowledge is the specific technical contribution that textile engineers make to composite preform development, and it is not easily acquired by structural engineers who have not studied textile manufacturing.

Beyond composite preforms, the textile engineer's manufacturing process knowledge — the operating principles of shuttle looms, rapier looms, Jacquard looms, and multi-axis braiding machines; the yarn path geometry of various weave patterns; the tension control requirements for processing carbon fibre without fibre breakage; and the quality control methods for detecting weave defects in structural fabric — transfers directly to composite dry fibre preform manufacturing engineering. The engineer who has operated or configured a loom for technical textile production can configure the same loom (or its industrial analogue) for carbon fibre weaving with specific guidance on the additional handling precautions required for brittle carbon fibre.

The textile engineer's understanding of fibre properties — specific the relationship between fibre surface treatment (sizing chemistry) and the fibre-matrix interfacial bond quality in a composite, the effect of fibre volume fraction on composite mechanical properties, the role of fibre twist on effective modulus, and the distinction between fibre properties (Young's modulus of individual carbon filaments) and yarn properties (reduced effective modulus due to twist and crimp) — is directly applicable to composite materials and process engineering in aerospace.

For protective systems and cabin safety, the textile engineer's understanding of flammability behaviour, heat transmission through layered textile systems, and tear resistance and seam integrity under dynamic loading are the specific competencies that make a textile engineer immediately valuable in cabin interior fire safety testing and protective system design.

The Specific Gaps — What Does Not Transfer Without Work

The aerospace regulatory framework for textile-containing products. The FAR 25.853 / CS-25.853 flammability requirements for cabin interior materials are the most immediately relevant regulatory framework for textile engineers entering aerospace cabin safety roles, but they are not the only regulatory dimension of their work. A seat cover that passes the FAR 25.853 vertical burn test and the heat release rate requirement of Part IV must also be assessed under the seat certification process (FAA TSO-C127 for aircraft seats), which considers the structural integrity of the textile under emergency loading conditions. An evacuation slide's textile envelope must be certified to the requirements of TSO-C69 (Emergency Evacuation Slides, Ramps, and Slide/Rafts). A parachute must be certified under TSO-C23 (personnel parachutes). Understanding these TSO certification processes — what they require in terms of qualification testing, analysis, and documentation — is the regulatory framework knowledge that textile engineers entering aerospace protective systems and cabin safety roles must have.

The structural certification framework. Textile engineers who target composite preform engineering — the most technically rewarding and most directly aligned application of their skills — must understand how the preform they design is integrated into a composite structural certification programme. The building block test approach (EASA AC 20-107B), the design allowables methodology (CMH-17), and the structural analysis methods (classical laminate theory for 2D composites, finite element analysis for complex geometries) are the structural engineering framework that determines whether the preform the textile engineer designs produces a part that meets the structural requirements of the aircraft. The textile engineer who knows the fabric

architecture but not the structural analysis chain from preform design to certified allowable is producing a preform that structural engineers must evaluate — not a complete engineering solution.

The failure mode vocabulary of safety-critical textile applications. Textile failure in a fashion or consumer context is measured in washing cycles and pilling resistance. Textile failure in an aerospace context is measured by the probability that the evacuation slide inflates correctly in 6 seconds, the probability that the parachute canopy deploys without an inversion, and the probability that the seat cover does not ignite and propagate flame to the seat foam under the FAR 25.853 vertical burn test. The aerospace textile engineer's failure analysis framework is fundamentally safety-consequence-driven, and the transition from consumer or industrial textile quality to aviation safety-critical textile quality requires a specific attitudinal and methodological shift: every textile design decision must be evaluated against its consequence if the textile fails in service.

The aerospace materials qualification framework. The textile engineer who selects a yarn for a technical application typically specifies it by commercial designation, tests it to verify it meets the required properties, and uses it. In aerospace, a material used in a safety-critical application must be qualified to a specification — an AMS specification, a DPAS specification for government programmes, or an OEM specification — with a documented qualification test programme, a certificate of conformance from the supplier, and a traceability chain from the raw material to the finished part. The yarn chemistry that determines fire resistance, the sizing formulation that determines fibre-matrix adhesion, and the fabric construction that determines preform permeability must all be specified, qualified, and controlled under a formal quality management system. The transition from commercial textile supply chain to aerospace supply chain represents a significant quality assurance step-change.

Composite structural analysis — classical laminate theory and the textile-specific modifications. Classical laminate theory, as taught in aerospace composites courses, is developed for unidirectional tape laminates — each ply has perfectly aligned fibres with zero crimp. Woven fabric composites require modifications to CLT because the crimp of the fibres in the weave introduces a reduction in the effective in-plane stiffness that is not captured by the standard CLT formulation. Understanding how the undulation angle of the crimp geometry in a plain weave, twill weave, or satin weave modifies the effective modulus of the resulting composite — and how this is accounted for in the design analysis — is the structural analysis adaptation that textile engineers targeting composite preform engineering must develop.

Liquid composite moulding processes — the processing context of dry fibre preforms. The dry fibre preform that the textile engineer designs is the starting point for a liquid composite moulding process — resin transfer moulding (RTM), vacuum-assisted resin infusion (VARI/VARTM), or resin film infusion (RFI). The preform architecture directly governs the resin flow behaviour during infusion: the in-plane permeability (how easily resin flows along the fabric plane) and the through-thickness permeability (how easily resin flows through the preform depth) determine whether the mould fills completely without dry spots or voids. Understanding how to characterise preform permeability (the Darcy permeability tensor for anisotropic fibrous media), how the weave architecture affects in-plane and through-thickness permeability (tightly woven fabrics have low in-plane permeability; loosely woven fabrics have high in-plane permeability), and what the consequences of incorrect permeability characterisation are for part quality (dry spots, voiding, fibre washing) is the process engineering context that the textile engineer designing preforms for LCM processes must have.

By Career Stage

Recent Textile Engineering Graduate

The recent textile engineering graduate has strong fibre science, fabric construction, and manufacturing process knowledge, with the least awareness of aerospace application context. The preparation task is specific: identify the target aerospace application domain (composite preforms, cabin safety, protective systems, or spacecraft soft goods), engage with that domain's regulatory framework (building block test approach for composites, FAR 25.853 for cabin safety, TSO-C69 for evacuation systems), and build at least one demonstrable connection between a textile engineering concept they know well and the specific aerospace engineering consequence it produces.

Mid-Career Textile Engineer (5–15 years)

The mid-career textile engineer has production experience, supplier management, and quality system knowledge that aerospace values in manufacturing and process engineering roles. The transition task is to con-

nect this experience to the aerospace quality framework — specifically to AS9100 quality management and NADCAP accreditation requirements for composite fabrication and materials testing, and to the specific aerospace material qualification pathway that governs the aerospace supply chain. The mid-career engineer who has managed a textile production facility to ISO 9001 is close to AS9100; the specific aerospace additions must be explicitly addressed.

Senior Textile Engineer / Technical Lead

The senior textile engineer who has led product development, managed qualification programmes, or developed technical specifications has the most transferable senior competencies — but must specifically demonstrate the regulatory maturity to own a qualification programme with airworthiness consequences. The senior interview will probe whether the candidate can navigate the aerospace certification framework with authority — whether they understand what it takes to get a novel composite preform architecture from design through qualification test programme to approved design allowable.

Domain Mapping — Where Your Textile Background Fits Best

Weaving and Fabric Architecture → Composite Dry Fibre Preform Engineering

Weave pattern geometry, areal weight, cover factor, drapability, yarn architecture — transfer directly. The aerospace addition: preform permeability for LCM processing, crimp modification to CLT stiffness predictions, CMH-17 allowables development for woven composites, and NADCAP composite fabrication requirements.

Braiding → Composite Braided Preform Engineering, Rope and Cable Systems

Braid angle, interlacing geometry, mandrel coverage, 2D and 3D braiding — transfer directly. The aerospace addition: biaxial and triaxial braid composite analysis (the braid angle determines the stiffness anisotropy), damage tolerance advantages of braided versus woven architectures, and for rope/cable: aviation specifications for restraint and recovery systems (MIL-DTL-83420 for wire rope substitute applications).

Technical Textiles / Coated Fabrics → Evacuation Slides, Life Rafts, Parachute Canopies

Coated fabric mechanics, seam design, pressure containment testing, dynamic load response — transfer directly. The aerospace addition: TSO certification processes (TSO-C69 for evacuation slides, TSO-C23 for parachutes), military specifications for parachute materials (MIL-C-7020 for nylon canopy fabric), and the specific dynamic deployment testing that characterises protective system performance.

Flammability and Fire-Resistant Textiles → Cabin Interior Fire Safety Engineering

Flame retardant mechanisms, limiting oxygen index, vertical burn testing, heat release rate characterisation — transfer directly. The aerospace addition: FAR 25.853 / CS-25.853 test methods and acceptance criteria (specifically the 12-second and 60-second vertical burn, the radiant heat test, and the smoke and toxicity requirements for large aircraft), the OSU heat release calorimeter test method, and the certification basis for new cabin interior materials including the aircraft-specific compliance demonstration pathway.

Filtration Textiles → Hydraulic System Filters, Fuel Filters, Cabin Air Filtration

Pore size distribution, filtration efficiency, dirt holding capacity, pressure drop characterisation — transfer directly. The aerospace addition: SAE AS755 for aircraft fuel system filters, AS1399 for hydraulic system filter elements, the specific chemical compatibility requirements for Skydrol hydraulic fluid contact, and the qualification test requirements for a new filter element as a component of a Part 25 aircraft system.

Yarns and Fibre Science → Composite Fibre Selection, Sizing Chemistry for Aerospace

Fibre tensile and compressive properties, fibre surface chemistry, sizing formulation and its effect on fibre-matrix interface — transfer directly. The aerospace addition: AMS 3892 and AMS 3894 for qualified carbon fibre roving, the sizing chemistry compatibility requirements for aerospace resin systems, and the qualification test programme for a new carbon fibre source in an approved composite prepreg system.

Knitting and Non-Wovens → Spacecraft Thermal Blankets (MLI), Airbag Landing Systems

Knit construction geometry, fibre lofting behaviour, compression-thickness relationships — transfer in modified form. The aerospace addition: MLI thermal performance analysis (the number of layers, the spacer fabric emissivity, and the contact conductance at the layer interfaces determine the effective insulation value), and for airbag systems, the dynamic energy absorption requirement under the specific landing deceleration profile.

PART II: 20 SPECIFIC PREP TASKS FOR THE TRANSITION

FOUNDATIONAL — DO THESE FIRST

1. Understand the composite preform design space — weave, braid, and 3D architectures and what each contributes structurally

5–6 hours

Skill: The textile-aerospace interface that is the most direct and most valuable technical connection

Why the field cares: Composite preform architecture is the primary design variable through which the textile engineer contributes to aerospace composite structures — and it a design variable that most composite structural engineers have not studied in depth because it is at the interface of their discipline and textile science. Plain weave produces the lowest crimp modulus reduction (approximately 5–10% versus UD tape) but the highest drapability (equal biaxial bending compliance). Satin weave produces higher drapability than plain weave (fewer crimp interlacings per unit area), better in-plane stiffness, and better surface quality (longer float lengths produce a smoother surface), but poorer stability during handling. 3D woven and 3D braided architectures introduce z-direction fibres that prevent interlaminar delamination — the primary failure mode of conventional 2D laminated composites under impact and out-of-plane loading — at the cost of reduced in-plane properties compared to 2D counterparts (the z-fibres occupy volume that would otherwise contain in-plane reinforcement). Understanding these trade-offs — and being able to explain to a structural engineer why a specific preform architecture is appropriate for a specific load case and fabrication method — is the technical communication competency that makes the textile engineer valuable in a composite development team.

Done signal: You can describe the structural consequence of fabric crimp on composite in-plane modulus — specifically that the fibre undulation in a woven fabric introduces a bending component to what should be a pure axial load, reducing the effective tensile modulus below the UD-tape value by an amount that depends on the undulation angle and can be estimated by the crimped fibre model. You can explain why a satin weave has better drapability than a plain weave in terms of the normalised bending rigidity per unit area. You can describe what a 3D woven preform provides that a 2D laminate does not — the through-thickness reinforcement that arrests delamination initiation — and identify one aerospace application where this advantage is specifically required (engine fan blades, thick composite fittings, or composite fuselage frames where out-of-plane loading is significant).

2. Understand liquid composite moulding processes — RTM, VARTM, and how preform architecture governs processing

4–5 hours

Skill: The manufacturing process context within which dry fibre preforms are used

Why the field cares: The dry fibre preform that a textile engineer designs is processed through a liquid composite moulding technique — resin transfer moulding, vacuum-assisted resin infusion, or resin film infusion — in which a liquid thermosetting resin is infused into the preform, cured under temperature and pressure, and removed from the mould as a finished structural part. The preform's permeability — how easily the resin flows through the fibre bed — is the critical processing parameter that the textile engineer controls through their choice of fibre architecture. A preform with very low permeability will be difficult to fully infuse, producing dry spots and voids in the finished part. A preform with non-uniform permeability will infuse unevenly, producing porosity gradients across the part. Understanding Darcy's law for flow through porous media ($Q = -kA/\mu \times dP/dx$, where k is the permeability, μ is the resin viscosity, and dP/dx is the pressure gradient), how to measure in-plane and through-thickness permeability experimentally, and how the weave architecture affects each permeability component (tighter weave = lower in-plane permeability; less tortuous through-thickness path = higher through-thickness permeability) is the process engineering knowledge that connects textile architecture design to manufacturing outcome.

Done signal: You can state Darcy's law for resin flow through a fibrous preform and explain what each term represents physically. You can describe how in-plane permeability is measured experi-

4–5 hours

mentally (the linear race tracking test: measure the flow front position as a function of time under a known inlet pressure, fit the data to Darcy's law to extract the permeability coefficient). You can explain what a dry spot is in an LCM part, what the consequence is for structural properties (stress concentration at the void, reduced interlaminar shear strength in the vicinity), and what preform design changes would reduce the risk of dry spot formation at a specific location.

5–6 hours

3. Understand FAR 25.853 / CS-25.853 — the flammability requirements for cabin interior materials

Skill: The primary regulatory framework for textile-containing products in commercial aircraft interiors

Why the field cares: Every textile product inside a commercial aircraft cabin — seat covers, carpets, curtains, wall panels, ceiling panels, and overhead bin liners — must meet the flammability requirements of FAR 25.853 (US) and CS-25.853 (Europe). These regulations specify the specific test methods and acceptance criteria for each material class, and the aircraft manufacturer must demonstrate compliance for every material in the cabin as part of the aircraft's type certificate. The vertical flame test (Appendix F, Part I — 12-second and 60-second exposure, measurement of burn length and flame propagation time) is the baseline requirement for most cabin materials. The heat release rate test (Appendix F, Part IV — the Ohio State University calorimeter, maximum heat release rate of 65 kW/m² and total heat release of 65 kW·min/m²) applies to large-area materials (seat cushions, wall panels, cargo compartment lining). The smoke density test (Appendix F, Part V) applies to materials in areas inaccessible to passengers. Understanding what each test method measures, what its acceptance criterion is, and what material modification strategies (intumescent coatings, inherently flame-resistant fibre systems, back-coating with FR treatment) achieve compliance are the cabin fire safety regulatory competencies that cabin interior engineering roles require from their first day.

Done signal: You can describe the FAR 25.853 Appendix F Part I vertical burn test — the test specimen orientation, the burner application time (12 seconds for the horizontal test position, 12 seconds for the vertical test position), the acceptance criteria (maximum average burn length of 6 inches for the 12-second test, with specific self-extinguishing time criteria), and what the test distinguishes (self-extinguishing materials from materials that sustain combustion after the ignition source is removed). You can explain what the OSU heat release rate test measures (the time-integrated rate of heat release from a material sample under a radiant heat flux of 35 kW/m²), why it is more relevant to cabin fire severity than the vertical burn test (heat release rate determines how quickly a cabin fire develops, which is what limits evacuation time), and what the FR modification strategy for a standard upholstery polyester would be to achieve compliance.

4–5 hours

4. Understand the TSO certification process for protective textiles — evacuation slides, parachutes, and restraints

Skill: The regulatory pathway for safety-critical textile products

Why the field cares: Technical Standard Order (TSO) authorisation is the FAA's certification for articles — components, equipment, or materials — that are designed and produced to meet minimum performance standards. For textile-containing safety-critical products: TSO-C69 covers emergency evacuation slides, ramps, and slide/rafts; TSO-C23 covers personnel parachutes; TSO-C114 covers torso restraint systems; and TSO-C127 applies to aircraft seats, which must meet dynamic testing requirements that include the textile seat cover's contribution to occupant protection. Understanding the TSO authorisation process — what minimum performance standards the product must meet, what testing is required to demonstrate compliance, what the documentation package for a TSO authorisation contains (the TSO Design Approval, the Statement of Conformance, the Manufacturer's Data Sheet), and how the TSO relates to the aircraft's type certificate — is the regulatory process knowledge that protective systems engineers and cabin safety engineers must have. The textile engineer who designs a technically excellent evacuation slide but does not understand the TSO-C69 compliance demonstration process cannot take their design to certification.

4–5 hours

Done signal: You can describe the TSO authorisation process — what the applicant submits (the TSO Design Approval application identifying the minimum performance standard, the test data demonstrating compliance, and the quality system documentation), what the FAA reviews (whether the minimum performance standard has been correctly identified and met), and what the TSO authorisation grants (the authority to manufacture the article in accordance with the approved design data). You can state what TSO-C69 requires for an emergency evacuation slide — the inflation time requirement (full inflation within 10 seconds from a 15°F ambient to 110°F ambient temperature range), the load capacity requirement (the design occupant load plus the dynamic test factor), and what the slide fabric must demonstrate in terms of tear strength and pressure retention after specific abuse testing scenarios.

4–5 hours

5. Understand the building block test approach for woven composite certification — how preform architecture affects the test programme scope

Skill: The certification methodology that connects preform design to structural allowables

Why the field cares: The composite preform architecture that a textile engineer designs must be qualified through the building block test approach before it can be used in a certified structure. The test programme differs from that for UD tape composites because woven and braided composites have additional architectural variables (weave pattern, braid angle, areal weight, yarn crimp) that affect mechanical properties and that must be bracketed by the test programme. The CMH-17 Volume 2 guidance for textile composites describes what additional tests are required relative to the UD tape test programme: the effect of weave pattern on tension, compression, and shear properties must be characterised; the effect of fabric thickness and areal weight on through-thickness properties must be addressed; and for 3D woven and braided architectures, the delamination resistance characterisation that demonstrates the z-fibre contribution to damage tolerance must be included. Understanding how the preform architecture determines the scope of the qualification test programme — and therefore the cost and schedule of introducing a new preform architecture into a certified structure — is the engineering judgement that makes a textile engineer valuable in composite certification discussions.

Done signal: You can describe how the CMH-17 building block test approach is modified for woven composite materials relative to UD tape — specifically what additional architectural variables must be bracketed by the test programme (weave pattern, fibre volume fraction range, fabric areal weight range), what additional test types are required (mode I and mode II interlaminar fracture toughness to characterise delamination resistance, through-thickness tension and shear to characterise out-of-plane loading capability), and why environmental conditioning (wet/hot) may have a different effect on a woven composite than on a UD tape composite (the resin-dominated through-thickness properties that are critical in 3D architectures are more sensitive to moisture than the fibre-dominated in-plane properties).

4–5 hours

6. Learn the AS9100 and NADCAP quality requirements for composite fabrication — how they differ from ISO 9001 and standard textile quality systems

Skill: The aerospace quality framework that governs composite preform manufacturing

Why the field cares: Textile manufacturers who supply composite reinforcement fabrics for aerospace use must operate under NADCAP-accredited composite fabrication procedures, and their customers — the composite part manufacturers — must also operate under NADCAP accreditation for the moulding, cure, and inspection processes. The NADCAP audit criteria for composite fabrication (checklist AC7118) cover: the cure cycle qualification (demonstrating that the specified temperature and pressure profiles are achieved and controlled within tolerance at every point in the cure cycle), the tool cleanliness and release agent application (which affect the composite surface quality and the validity of the cure cycle), the first-article inspection scope (which for textile composites includes verification of fibre orientation angles, areal weight, and fabric edge quality in addition to the standard dimensional and visual inspection), and the personnel quali-

4–5 hours

fication requirements (lay-up personnel must demonstrate competency through qualification lay-up that are inspected for fibre orientation accuracy and compaction quality). Understanding what NADCAP compliance requirements are for composite fabrication that have no equivalent in the target), the controlled environment requirement for lay-up areas (temperature, humidity, and contamination controls that prevent moisture uptake in hygroscopic carbon fibre sizing before cure), and the ultraviolet light inspection requirement for detecting contamination on composite surfaces that would prevent adhesive bonding or subsequent ply adhesion. You can explain what the consequence would be of supplying a composite reinforcement fabric from a non-NADCAP source to an aerospace prime contractor with an AS9100 quality system — specifically that the prime contractor cannot use the non-NADCAP fabric in a flight-critical application regardless of its technical quality.

5–7 hours

7. Build competence in the structural analysis of woven composites — the crimp correction to CLT and the failure criteria

Skill: The structural mechanics of woven composites — the analysis framework that connects preform architecture to structural performance

Why the field cares: The textile engineer targeting composite preform engineering must understand how their preform architecture affects the structural analysis of the resulting composite. Classical laminate theory (CLT) as applied to UD tape laminates predicts the stiffness and strength of a laminate from the properties of individual plies. For woven fabric composites, CLT must be modified to account for the effect of fibre crimp on effective ply modulus: the undulation angle of the fibre as it crosses the weave crimp reduces the effective in-plane tensile modulus below the value predicted by the rule of mixtures. The crimp correction model (Naik and Shembekar, or equivalent) predicts the modulus reduction as a function of the undulation angle, which depends on the weave pattern and the fabric areal weight. Understanding this correction, being able to estimate the crimp angle from the fabric geometry, and applying the corrected modulus values in the CLT analysis framework is the structural analysis knowledge that connects the textile engineer's fabric design to the structural engineer's analysis output.

Done signal: You can explain why the in-plane tensile modulus of a plain weave carbon/epoxy composite is lower than that of an equivalent UD tape laminate — specifically that the fibre crimp introduces a transverse stress component that reduces the effective axial load-carrying capacity of the fibre. You can estimate the crimp angle from the fabric geometry (the undulation height divided by the half-wavelength gives the approximate maximum crimp angle) and describe qualitatively how the modulus reduction scales with crimp angle. You can describe what the Tsai-Wu failure criterion requires for application at the ply level in a woven composite — specifically whether the interaction terms must be characterised separately for woven composites relative to UD tape, because the failure modes (fibre-dominated tension failure versus matrix-dominated compression or shear failure) remain the same.

3–4 hours

8. Understand the fibre-matrix interface in aerospace composites — sizing chemistry and its effect on mechanical properties

Skill: The fibre science most directly relevant to composite preform engineering

Why the field cares: The sizing applied to carbon fibre filaments during manufacturing determines the quality of the fibre-matrix interface in the finished composite — and the fibre-matrix interface quality determines the interlaminar shear strength, the transverse tensile strength, and the compression strength of the composite. Sizing is a dilute aqueous solution of a polymeric coupling agent (typically an epoxy-compatible film former) applied to the fibre surface during the manufacturing process to protect the fibre from abrasion during weaving and to promote adhesion to the matrix resin. The sizing chemistry must be compatible with the resin system used in the composite: a sizing designed for epoxy compatibility will not perform well with a polyimide matrix used in high-temperature applications. The effect of weaving on sizing integrity — whether the mechanical abra-

3–4 hours

sion during the weaving process degrades the sizing and reduces the fibre-matrix interface quality — is a specific concern for high-performance aerospace carbon fibre fabrics that textile engineers must understand. Weaving carbon fibre is substantially more demanding than weaving glass or aramid fibre because carbon fibre is brittle and has very low bending radius before breakage — the weaving parameters (tension, shedding speed, reed geometry) must be specifically adapted for carbon fibre to prevent fibre breakage and sizing damage.

Done signal: You can explain what sizing is and what its two functions are in a composite preform (fibre protection during manufacturing and fibre-matrix adhesion promotion), describe what “sizing compatibility” means in the context of a specific carbon fibre/resin combination (the sizing must be chemically compatible with the resin to form a strong interfacial bond rather than acting as a release layer), and identify what interlaminar shear strength (ILSS) measures (the shear strength at the fibre-matrix interface and in the matrix between adjacent plies — a low ILSS indicates poor fibre-matrix bonding or poor matrix quality). You can describe what happens to ILSS when a fabric is woven from carbon fibre with damaged sizing, and explain what the consequence is for the structural performance of the resulting composite.

2–3 hours

9. Prepare behavioural answers that connect textile decisions to safety and structural consequences

Skill: The narrative translation that connects textile engineering expertise to aerospace-level engineering consequences

Why the field cares: Aerospace interviewers hiring textile engineers must assess whether the candidate understands that their textile decisions have consequences that extend beyond material performance in isolation to aircraft safety or structural integrity. The textile engineer who says “I developed a fire-resistant polyester face fabric that passed the vertical burn test” has described a technical achievement. The textile engineer who says “I developed a fire-resistant polyester face fabric that passed the FAR 25.853 Appendix F Part I 12-second vertical burn test, demonstrated a burn length below the 4-inch acceptance criterion at the minimum ambient temperature condition, and assessed that the FR additive concentration was sufficient to maintain compliance after the 10-wash durability requirement that the airline’s seat cover cleaning programme imposes — I also considered whether the FR additive’s toxicity under combustion was within the cabin smoke and toxicity limits under the applicable AC 25.853-1 guidance” has demonstrated the complete chain from textile technology decision to aerospace regulatory consequence. Every aerospace textile interview will probe whether the candidate can make this chain explicit.

Done signal: Four STAR stories covering: a textile material or construction decision with a specific performance consequence (quantified); a textile quality problem you identified and resolved, with specific connection to the product’s intended function; a situation where a textile design trade-off was required (performance versus cost, drapability versus structural performance, FR treatment versus durability) and how you resolved it; and a textile failure or nonconformance that you investigated, with specific connection to the root cause in the textile’s structure or processing. At least two stories demonstrate the chain from a textile observation to a structural, safety, or regulatory consequence.

15–25 hours

10. Build a portfolio project demonstrating aerospace-domain textile engagement

Skill: The technical demonstration that distinguishes genuine aerospace preparation from generic textile interest

Why the field cares: Textile engineers have an unusual portfolio project opportunity: most of the relevant aerospace textile work is physical — fabric characterisation, flammability testing, preform construction — and accessible at relatively low cost if the textile engineering department or a technical textiles supplier can provide material access. Accessible projects include: a comparative permeability characterisation of three different weave architectures (plain, 2×2 twill, 5-harness satin) in the same carbon fibre areal weight, measuring the in-plane permeability by the linear race tracking method and the through-thickness permeability by the through-thickness flow method, and com-

15–25 hours

paring the results against the Kozeny-Carman prediction; a fire resistance testing programme on candidate materials for a notional aircraft interior application, following the FAR 25.853 Appendix F Part I vertical burn methodology at reduced scale in a properly controlled laboratory setting, and comparing the results with the published acceptance criteria; or a crimp angle measurement programme on woven carbon fabric samples of different weave patterns, using optical microscopy of polished cross-sections to measure the fibre undulation angle, and comparing the measured modulus reduction against the crimp correction prediction. Each of these is achievable in a university textile engineering laboratory setting and produces specific technical content for the interview.

Done signal: You have completed a portfolio project in your target aerospace textile application domain — composite preforms, fire-resistant cabin materials, or protective systems — that produces a specific, documented technical output with clearly stated methods, results, and engineering conclusions. The documentation includes the aerospace regulatory or engineering context (what the test was addressing and why it matters in an aerospace application), the specific test method or analysis used (referenced to a standard, an academic paper, or an aerospace specification), and a comparison against relevant aerospace acceptance criteria or predictions. You can present it in 10 minutes.

ROLE-SPECIFIC — TARGET AFTER FOUNDATIONAL TASKS

4–5 hours

11. For composite preform roles: understand 3D weaving and 3D braiding — architecture, structural advantage, and manufacturing constraints

Skill: The advanced preform architectures that offer the greatest structural advantage for aerospace primary structure

Why the field cares: 3D woven and 3D braided composites are the preform architectures that most directly exploit the unique structural capability of textile reinforcement — the ability to place fibres in the through-thickness direction that conventional 2D laminates cannot provide. 3D woven preforms use warp and weft yarns in the in-plane directions, with binder yarns that thread through the thickness to connect multiple layers into a single integrated preform. The binder yarn architecture (through-the-thickness orthogonal, layer-to-layer, or angle-interlock) determines the balance between in-plane stiffness (reduced by the binder yarn that displaces in-plane fibres) and delamination resistance (increased by the binder yarn that arrests crack growth between layers). 3D braiding (both Cartesian and rotary braiding) produces a preform in which all yarns are interlaced in three dimensions, providing excellent damage tolerance but reduced in-plane stiffness. Manufacturing constraints for 3D woven preforms are severe — the Jacquard loom programming required for complex 3D architectures is substantially more demanding than for 2D fabrics, and the preform shape is constrained to prismatic forms that can be woven on a flat bed and subsequently shaped. Understanding these trade-offs is the advanced preform design knowledge that employers building fan blades, thick fittings, or joints for composite primary structure require.

Done signal: You can describe the three main categories of 3D woven architecture (orthogonal, layer-to-layer, and angle-interlock) and explain what each provides in terms of the trade between in-plane stiffness and through-thickness reinforcement effectiveness. You can explain why orthogonal 3D weaving has the highest through-thickness strength but the lowest in-plane properties of the three architectures (the binder yarns are oriented purely in the z-direction, providing maximum through-thickness reinforcement but displacing the maximum volume of in-plane fibres). You can describe one specific aerospace application for a 3D woven preform — a composite fan blade for a turbofan engine, where the through-thickness reinforcement from the 3D weave improves bird strike damage containment relative to a 2D tape laminate blade.

4–5 hours

12. For protective systems roles: understand evacuation slide design requirements — inflation systems, fabric envelope mechanics, and dynamic load certification

Skill: The largest and most common aerospace technical textile product — with specific airworthiness requirements

4–5 hours

Why the field cares: Emergency evacuation slides are the most safety-critical technical textile product in commercial aviation: their function — to evacuate all passengers from a maximum-size commercial aircraft in 90 seconds — is directly life-critical. The slide's textile envelope is a coated fabric (typically nylon 6,6 or nylon 6,12 coated with polyurethane or neoprene) that must contain the pressurised inflation gas, provide structural support for evacuating passengers sliding down its inclined surface, and survive the inflation loads without seam or fabric failure. The coated fabric must meet the tear and tensile strength requirements of the applicable specification (MIL-C-7439 or AMS 3806 for coated fabrics in inflatable evacuation systems), must retain its properties through the temperature range specified in TSO-C69 (-65°F to 160°F), and must be hermetic to prevent nitrogen loss through the fabric (the inflation gas must be retained for at least 30 minutes after full inflation to allow extended evacuation in a ditching scenario). Understanding how the fabric's coating system determines the impermeability (the coating must bridge all interstices in the base fabric without cracking under cyclic flexing during the inflation and use cycles), and what the consequence of coating failure is (gas leakage, reduced inflation pressure, and potential collapse of the slide during use) is the protective systems textile engineering knowledge that evacuation slide manufacturers require.

Done signal: You can describe the primary performance requirements for an evacuation slide fabric — permeability, tear strength at room temperature and at temperature extremes, tensile strength in the warp and weft directions, coating adhesion under the inflation flexing cycle, and resistance to UV degradation and hydraulic fluid contamination. You can explain what a seam failure during inflation means for the slide's function (a seam failure during inflation produces a partial deployment that cannot support the passenger load, and may cause the slide to collapse during an evacuation). You can describe how the seam design (stitched, bonded, or radio-frequency welded) affects the failure mode and the seam efficiency (the ratio of seam strength to base fabric strength).

5–6 hours

13. For spacecraft soft goods roles: understand space suit pressure garment construction — the specific textile layers and their functions

Skill: The most technically demanding textile application in any industry — space suit design

Why the field cares: The extravehicular mobility unit (EMU) or equivalent space suit is a pressurised garment system that must simultaneously contain a pressure of approximately 4.3 psi, provide thermal regulation from -250°F to +250°F, protect against micrometeorite impact, resist UV and atomic oxygen degradation in the LEO environment, allow sufficient mobility for EVA tasks, and operate for up to 8 hours without maintenance. It is the most technically demanding soft goods engineering application that exists, and it is currently undergoing the most significant redesign in decades as NASA and its commercial partners develop the Exploration Extravehicular Mobility Unit (xEMU) for Artemis lunar operations and commercial EVA suits for orbital station operations. The space suit pressure garment comprises multiple technical textile layers: the Thermal Micrometeoroid Garment (TMG) outer layer uses multiple layers of aluminised Mylar for thermal insulation with a woven Kevlar/Dacron restraint layer for micrometeorite protection; the pressure bladder is a seamless or minimally-seamed urethane-coated nylon 6,6 garment that contains the pressure without stretching; the restraint layer over the bladder is a woven Dacron structure that limits the bladder expansion under pressure to maintain the garment shape; and the comfort liner in contact with the suit occupant is typically a knit polyester or polypropylene wicking fabric that manages perspiration. Understanding the function of each layer and the textile-specific engineering constraints that apply to each is the space suit engineering literacy that soft goods developers at ILC Dover, David Clark Company, and Final Frontier Design require.

Done signal: You can describe the function of each of the four primary space suit textile layers (TMG, bladder, restraint, liner), the specific textile structure used for each (the aluminised Mylar layer count and spacing, the urethane bladder construction, the Dacron restraint weave architecture, the wicking knit structure), and the primary failure mode of each that the textile design must manage (TMG: coating delamination from thermal cycling; bladder: seam leak from cyclic pressure flexing; restraint: creep under sustained pressure load; liner: moisture management failure). You can explain what the mobility requirement means for the restraint layer design — specifically that the garment must allow the occupant to flex joints against the pressure load, which requires the restraint architecture to

have specific in-plane compliance in the bending direction of each joint that the bladder does not inherently provide.

5–6 hours

14. For parachute and deceleration roles: understand parachute materials specifications and canopy design principles

4–5 hours

Skill: The structural textile engineering of deceleration systems

Why the field cares: Parachutes for military and space applications are engineered systems that must deploy reliably from a packed state in fractions of a second, sustain opening loads of tens of thousands of pounds, and maintain structural integrity through a specified number of use cycles. The canopy fabric — typically a 1.1 oz/yd² nylon 6,6 ripstop (MIL-C-7020 for Type III canopy fabric) — must have specific tear strength, porosity, and dimensional stability under load that are set by the canopy aerodynamic design. The porosity of the canopy fabric determines the stability of the parachute at steady descent (a porous canopy has lower opening loads and better stability than an impermeable canopy, at the cost of a higher descent rate). The suspension lines — typically 550 lb rated nylon cord for personnel parachutes — must have a specific elongation characteristic that limits the deceleration impulse to within the human tolerance limit (approximately 15g for a well-designed personnel parachute). Understanding the relationship between fabric porosity, canopy stability, and descent rate is the parachute aerodynamics knowledge that connects the textile engineer's fabric characterisation competency to the parachute system performance requirements.

Done signal: You can describe the porosity requirement for a personnel parachute canopy fabric and explain why a specific porosity range (typically 3–10 CFM/ft² at 0.5 inches water pressure differential for a standard round canopy) is required — specifically that insufficient porosity causes oscillation instability (the canopy breathes and spills air unevenly, causing the pendulum-like oscillation) while excess porosity reduces the aerodynamic drag coefficient and increases descent rate beyond the safe landing speed. You can describe what the MIL-C-7020 specification requires for a Type III canopy fabric in terms of fabric weight, tear strength, elongation, and porosity. You can explain what a ripstop weave provides relative to a plain weave — the heavier reinforcing yarn woven at regular intervals that arrests tear propagation if the fabric is punctured.

15. For filtration roles: understand the aerospace hydraulic and fuel system filter requirements

4–5 hours

Skill: The fluid system filtration engineering specific to aerospace applications

Why the field cares: Aircraft hydraulic systems and fuel systems use filter elements — porous textile structures — to remove particulate contamination that would cause valve sticking, actuator wear, or fuel system component damage. The filter element in an aerospace application must meet requirements that go substantially beyond industrial filtration: the filtration efficiency must be characterised by the Beta ratio (the ratio of particles upstream to particles downstream of the filter at a specified particle size, measured under the ISO 16889 multi-pass test method), the pressure drop across the element must stay within the system design limit across the full operating range of flow and temperature, the element must maintain its structural integrity when exposed to Skydrol hydraulic fluid (a phosphate ester fluid that is highly aggressive to many polymers), and the element must not contribute fibres, particulates, or soluble contamination to the fluid system. Understanding what the ISO 16889 Beta ratio test measures, what a Beta ratio of 1000 at 3 microns means for system cleanliness (99.9% filtration efficiency for particles ≥ 3 microns), and what the Skydrol compatibility test (MIL-PRF-5606 fluid compatibility test or equivalent) requires from the filter medium are the filtration engineering competencies that aerospace fluid system filter suppliers require.

Done signal: You can explain what the Beta ratio is — the filtration efficiency metric from the ISO 16889 multi-pass test — and calculate what a Beta₃ = 1000 means for the percentage filtration efficiency at 3-micron particle size (99.9% removal efficiency). You can describe what Skydrol is (a triaryl phosphate ester hydraulic fluid used in commercial aircraft for its fire resistance relative to mineral oil hydraulics), what makes it aggressive to textile filter media (the phosphate ester groups hydrolyse polymer ester bonds, degrading polyester and some polyamide media), and what media materials

are Skydrol-compatible (glass microfibre, sintered PTFE, PTFE membrane on a glass fibre substrate). You can describe what a filter element must demonstrate in the burst pressure test (the differential pressure at which the element fails mechanically, which must exceed the system maximum pressure by a specified safety margin).

4–5 hours

16. For MLI and thermal blanket roles: understand multi-layer insulation — layer count, spacer fabric function, and the spacecraft thermal environment

4–5 hours

Skill: The thermal management application for textile structures in the space environment

Why the field cares: Multi-layer insulation (MLI) is the standard passive thermal management system for spacecraft that must maintain component temperatures within their operating limits in the space environment. MLI consists of alternating layers of aluminised polyimide (Kapton) or aluminised PET (Mylar) film separated by low-conductance spacer layers — typically Dacron net fabric or silk net fabric. The thermal performance of MLI is dominated by radiation heat transfer between the aluminised layers (the emissivity of the aluminium coating determines the radiative coupling between adjacent layers) and by the contact conductance at the layer interfaces (the spacer fabric introduces a mechanical contact resistance that reduces the solid conduction heat flow between adjacent film layers). The spacer fabric selection is a balance between minimum contact conductance (favouring coarser, more open-structured fabrics that minimise the number and area of contact points) and minimum outgassing (the fabric material must have very low vapour pressure in vacuum to avoid contaminating sensitive optical surfaces). Understanding the thermal resistance contribution of each layer pair and the specific outgassing requirements for spacecraft materials (meeting the NASA MSFC-SPEC-1124 total mass loss and collected volatile condensable materials criteria) is the spacecraft thermal materials knowledge that MLI and thermal blanket engineering roles require.

Done signal: You can describe the MLI thermal performance model — explaining why the effective emittance of an MLI blanket decreases as the number of layers increases (each additional layer adds another radiation shield, reducing the net radiation exchange between the inner and outer boundaries), what determines the minimum effective emittance achievable (the aluminisation quality and the degree of edge and seam penetration), and what the typical effective emittance is for a 25-layer MLI blanket in vacuum (approximately 0.002–0.003). You can describe what the Dacron net spacer fabric provides that direct contact between aluminised film layers does not (thermal contact resistance that reduces solid conduction heat flow, which would otherwise dominate the heat transfer if the film layers contacted each other directly). You can explain what outgassing means in the spacecraft thermal blanket context and why it matters (outgassed molecules from the fabric condense on optical sensors, detectors, or mirror surfaces, reducing their optical performance).

17. Understand the aerospace supply chain quality requirements for textile raw materials — trace, certify, and qualify

3–4 hours

Skill: The material traceability and qualification framework specific to aerospace supply chain management

Why the field cares: The aerospace supply chain applies material traceability and qualification requirements to every raw material that enters a flight-critical product that have no equivalent in the commercial textile industry. Every spool of carbon fibre yarn, every roll of woven fabric, and every batch of sizing must be traceable to its manufacturer's Certificate of Conformance (CoC), which certifies that the material meets the applicable specification (AMS, MIL, or OEM specification) in every required property. The CoC is not just paperwork — it is the legal document that allows the receiving organisation to incorporate the material into a flight-critical product, and a missing or incorrect CoC may require re-testing of the entire lot or rejection of the material regardless of its technical quality. Additionally, new materials — a new carbon fibre type, a new fabric weave architecture, a new sizing formulation — must be qualified through a documented qualification test programme before they can be incorporated into a certified product.

Done signal: You can describe what an aerospace raw material Certificate of Conformance must con-

3–4 hours

tain (the material designation and revision, the specification number and revision, the specific test results that demonstrate conformance, the test methods used, the lot number, the date of manufacture, and the authorised signature of the certifying official), and explain what the receiving organisation must verify when they receive a CoC (that the material specification is the one called out in the design data, that the revision is current, that all required tests are reported, and that the results meet the acceptance criteria). You can describe what a material qualification programme for a new carbon fibre type in an approved composite system requires — the property characterisation scope, the building block test sequence, and the authority approval required before the new fibre can be used in a certified structure.

4–5 hours

18. Build quantitative estimation fluency for aerospace textile applications

Skill: First-principles estimation about textile performance in aerospace applications

Why the field cares: Senior aerospace textile engineers are expected to provide immediate preliminary assessments of textile performance questions — to estimate whether a candidate fabric can meet the permeability requirement for a specific LCM application before the measurement is made, to assess whether a proposed seam design has the required efficiency for the inflation load on an evacuation slide, or to estimate the number of MLI layers needed to achieve a specified effective emittance. These estimates are tractable from textile science fundamentals: estimating the in-plane permeability from the Kozeny-Carman equation using the fibre volume fraction and the hydraulic diameter; estimating the burn length in a vertical burn test from the flame propagation time and the material's LOI; or estimating the MLI effective emittance from the layer count and the aluminium emissivity. Senior engineers who can produce these estimates quickly are more valuable in real-time engineering discussions than those who require a full characterisation test before any assessment.

Done signal: You have worked through five aerospace textile estimation problems covering at least two of: in-plane preform permeability from Kozeny-Carman for a specified weave and fibre volume fraction; crimp angle and modulus reduction for a plain weave versus satin weave at the same areal weight; burn length estimation from LOI and vertical burn test conditions; MLI effective emittance from layer count and aluminium emissivity; and seam efficiency estimation from fabric tensile strength and seam stitch density for a candidate evacuation slide seam design. You state your assumptions before numbers, give order-of-magnitude answers with stated uncertainty, and identify the most load-bearing assumption in each.

POSITIONING AND PRESENTATION

2–3 hours

19. Prepare a specific answer to “Why aerospace — and what from your textile engineering background makes you ready for this role?”

Skill: The transition motivation that addresses the specific surprise aerospace interviewers feel when they see a textile engineering background

Why the field cares: Aerospace interviewers who see a textile engineering background on a CV have a specific initial uncertainty: they are not sure whether the candidate understands how their textile knowledge applies to aerospace, or whether they are making a generic “I want to work in aerospace” transition without understanding the specific technical connection. The answer must resolve this uncertainty immediately and specifically. The answer that says “I have always been interested in aerospace and I think my textile knowledge is transferable” does not resolve it. The answer that says “My research on braided composite preform permeability for resin transfer moulding processes maps directly to the composite aerostructure manufacturing challenge your engineering team is working on — specifically the preform architecture design for the next-generation composite rib for the wing leading edge structure, where the 3D braid angle and the through-thickness yarn volume fraction are the design variables that I can optimise for both structural performance and RTM processing quality” resolves it immediately. The specificity of the aerospace application — naming the part, the material, the process, and the textile design variable — is what makes the answer credible.

2–3 hours

Done signal: You have a 2-minute answer that: names the specific aerospace application domain and the specific textile engineering problem within it; explains precisely which of your textile competencies is most directly relevant and why; demonstrates that you understand the aerospace regulatory or structural engineering context in which your textile work would be evaluated; and connects the consequence of doing the textile engineering work correctly to a specific aircraft or spacecraft performance, safety, or certification outcome. The answer cannot be given by any generic aerospace engineering candidate — it requires textile engineering knowledge to make sense.

4–6 hours

20. Map your textile specialisation to the specific aerospace employers and roles where it is most competitive

Skill: The job market intelligence that focuses the transition on the highest-probability opportunities

Why the field cares: The aerospace textile job market is narrower than any other transition in this series — the roles are fewer, more specialised, and more geographically concentrated. Targeting matters more here than in any other transition. The composite preform specialist should target Albany Composites (the world's largest 3D woven composite preform manufacturer for aerospace — their LEAP engine fan blades are the most commercially deployed aerospace 3D woven composite component), Safran Composites Preforms, A&P Technology (braided composite preforms), Hexcel (their dry fabric division as distinct from their prepreg division), and the composite manufacturing groups at Airbus, Boeing, and GE Aviation. The cabin fire safety specialist should target Safran Seats, Collins Aerospace Cabin Interiors, and the certification engineering groups at major airlines that manage cabin interior approval programmes. The protective systems specialist should target Cobham Life Support, Survival Systems International, and the parachute system manufacturers (Airborne Systems, Pioneer Aerospace). The space suit specialist should target ILC Dover, David Clark Company, and NASA's space suit development programme. Applying broadly without this targeting is the most common failure mode for textile engineers attempting the aerospace transition, and it is the failure mode that this guide is specifically designed to prevent.

Done signal: You have identified at least five specific employers where your textile specialisation is directly relevant — verified against their published job descriptions, their product lines, and (where available) their technology development programmes. You have mapped the specific textile engineering skills that each employer requires to the specific skills you have and the specific gap tasks from this guide that you have completed. You have at least two active applications, each with a portfolio project specifically relevant to the employer's textile application domain and a cover letter that explicitly names the textile engineering problem the employer is working on and explains how your specific background addresses it.

SUMMARY REFERENCE TABLE

#	Task Description
1 All / Do first	Composite preform architecture — weave, braid, 3D architectures and structural trade-offs
2 Composite preform	Liquid composite moulding — RTM, VARTM, preform permeability and Darcy's law
3 Cabin safety / All	FAR 25.853 / CS-25.853 — vertical burn, heat release, smoke and toxicity requirements
4 Protective systems	TSO certification — TSO-C69 for slides, TSO-C23 for parachutes, authorisation process
5 Composite preform	Building block test approach for woven composites — architectural variables and test scope
6 Composite / All	AS9100 and NADCAP for composite fabrication — what textile QS lacks, cure cycle qualification
7 Composite preform	Structural analysis of woven composites — crimp correction to CLT, failure criteria
8 Composite preform	Fibre-matrix interface — sizing chemistry, ILSS, weaving effects on interface quality
9 All	STAR stories — textile decision to structural/safety consequence, investigation chain
10 All / Critical	Portfolio project — aerospace textile domain, regulatory context, specific output
11 Composite preform	3D weaving and braiding — orthogonal/layer-to-layer/angle-interlock, fan blade application
12 Protective systems	Evacuation slide design — coated fabric, inflation loads, TSO-C69 requirements
13 Spacecraft soft goods	Space suit pressure garment — TMG, bladder, restraint, liner layers and functions
14 Parachute systems	Parachute materials — MIL-C-7020 canopy fabric, porosity, suspension line specification
15 Filtration	Hydraulic/fuel system filter requirements — Beta ratio, Skydrol compatibility, burst pressure
16 Spacecraft thermal	Multi-layer insulation — layer count, spacer fabric, effective emittance, outgassing
17 All	Aerospace material CoC and qualification — traceability requirements, new material qualification
18	Quantitative estimation — permeability, crimp modulus, burn length, MLI emittance

#	Task Description
All	
19	“Why aerospace — what specifically from textile engineering?” — 2-minute targeted answer
All	
20	Textile specialisation to employer mapping — composite, cabin, protective, space, filtration
All	

THE STRUCTURAL REALITY OF THE TEXTILE ENGINEERING-TO-AEROSPACE TRANSITION

The structural reality of the textile engineering-to-aerospace transition: The textile engineer entering aerospace is doing something that most of the aerospace industry does not know needs doing — which is precisely why the opportunity is real. The composite structures group at Airbus, the evacuation systems manufacturer in Gosport, the space suit developer in Delaware, and the spacecraft thermal blanket manufacturer in Minnesota all employ engineers who understand aerospace systems, structures, and certification. What they often lack is someone who genuinely understands the textile — who knows that the crimp angle in a plain weave composite is 15% higher than in a 5-harness satin at the same areal weight and that this matters for the modulus prediction, who knows that the ripstop yarn count determines the arrest width of a tear propagating through the canopy fabric, and who can look at a fabric cross-section and identify from the sizing distribution whether the fibre-matrix interface will fail cohesively or adhesively.

This specific knowledge — the kind that comes from having studied textile manufacturing, fibre science, and fabric mechanics as a primary discipline — is genuinely rare in aerospace, and its rarity makes it specifically valuable rather than merely interesting. The composite structural engineer who knows classical laminate theory but who treats the fabric as a UD tape equivalent is making a systematic modelling error every time they compute the stiffness of a woven panel. The evacuation slide engineer who knows FAR 25.853 but who cannot specify the coating formulation for a polyurethane-coated nylon to meet the permeability and tear strength requirements simultaneously is dependent on their fabric supplier to make a decision they should be making. The space suit engineer who knows the pressure containment requirement but who cannot specify the bladder seam construction to achieve the required hermetic integrity over 8 hours of pressure cycling is accepting a structural risk that the textile engineer would manage as a daily engineering decision.

The transition’s challenge is visibility, not competence. The solution is targeting — identifying the specific employers where textile knowledge creates the gap that the candidate fills, rather than the much larger set of employers where the connection is too indirect to be immediately apparent. The employers in this guide’s target set — Albany Composites, ILC Dover, Airborne Systems, Cobham Life Support, the composite manufacturing groups at major OEMs — actively need what textile engineers bring. They are not being asked to make a leap of imagination. They are being asked to hire someone whose skills they recognise as directly relevant once the connection is made explicit.

Making that connection explicit — in the application letter, in the portfolio project, and in the interview room — is the entire task of this transition. The textile science is in place. The aerospace regulatory context is learnable, specific, and bounded. And the combination of the two, in the hands of someone who has genuinely engaged with both, is exactly the engineering profile that the most technically demanding textile applications in any industry require.

BREAK INTO AEROSPACE
